

Fan (Nils) Yang, Ph.D

Bldg. 10, Rm. B1D-723A, 10 Center Drive, Bethesda, MD 20892
Email: nilsyang106@gmail.com
Permanent Resident (Green Card)

Phone: 267-815-8421
Website: nilsyang.github.io

Employment

2023.01 - present Research Fellow, NINDS, NIH. Advisor: Jeff Duyn.
2021.05 - 2022.12 Postdoctoral Fellow, University of Maryland, Baltimore. Advisor: Ze Wang.
2018.12 - 2021.02 Postdoctoral Fellow, Georgetown University. Advisor: Xiong Jiang.

Education

2015.12 - 2018.11 Visiting Ph.D. Student, Department of Neurology & Center for Functional Neuroimaging, University of Pennsylvania. Advisor: Hengyi Rao & John Detre.
2014.09 - 2018.06 Ph.D., Cognitive Neuroscience, Sun Yat-Sen University (SYSU), Guangzhou, China. Advisor: Hengyi Rao.
2011.09 - 2014.06 M.S., Biological Psychology, Capital Normal University, Beijing, China
2006.09 - 2010.06 B.S., Chemistry & Clinical Psychology, Southwest University, Chongqing, China

Additional Training & Certificates

2023 Scientist Teaching Science (9-week online course). Instructor: Nik Barkauskas
2018 Summer School in Computational Sensory-Motor Neuroscience (CoSMo).
Coordinators: Drs. Gunnar Blohm, Paul Schrater, & Konrad Körding
2013 Level 2 Psychological Counselor, National Vocational Qualification of China
2010 Teacher Qualification Certificate for high school, Ministry of Education, China

Science Communication & Outreach

2014 - present Science communication on Zhihu (zhihu.com), a Chinese knowledge-sharing platform: sleep, neuroscience, and research methods for a general audience; 181,401 followers as of August 2026.

Representative Media Coverage

Too little sleep may jeopardize young kids' development, study finds. *The Washington Post*
<https://www.washingtonpost.com/health/2022/08/09/sleep-children-brain-development/>

Grant Activity

2026 NIH K01 application submitted June 2026

Honors, Awards & Fellowships

2024 Trainee Merit Reward, Sleep Meeting 2024
2023 - 2024 Performance Cash Award, NINDS
2018 Fellowship, Summer School in Computational Sensory-Motor Neuroscience
2015 International Program Fellowship for Ph.D. Candidates, SYSU
2013 Postgraduate Research Fellowship, Capital Normal University

Editorial Service

Ad Hoc Reviewer

BMC Neuroscience; Drug and Alcohol Dependence; Frontiers in Human Neuroscience; Frontiers in Neurology; Frontiers in Neuroimaging; Human Brain Mapping; Journal of Magnetic Resonance Imaging; Network Neuroscience; SLEEP; Psychological Medicine; Scientific Reports; Imaging Neuroscience; The Lancet Child & Adolescent Health; Sleep Medicine Reviews; PNAS.

Reviewing editor

Editorial Board of Neuroimaging for Cognitive Neuroscience, Frontiers in Neuroimaging

Teaching

2015 Teaching Assistant, Cognitive Psychology (in English), SYSU
2015 Teaching Assistant, Ecological Psychology, SYSU
2014 Teaching Assistant, Experimental Psychology (in English), SYSU

Mentoring

2023 - 2024 Co-mentor, NIH Postbaccalaureate Intramural Research Training Award (IRTA)
Mentee: Hadar B. Halivni, BA.
2019 - 2021 Co-mentor, Ph.D. candidate in Interdisciplinary Program in Neuroscience (IPN)
Mentee: Neke Nosor, MS.
2018 - 2021 Co-mentor, Ph.D. candidate in Interdisciplinary Program in Neuroscience (IPN)
Mentee: Shiva Hassanzadeh-Behbahani, BA.

Invited Talks & Conference Talks

2025.06 Increased intra-thalamic and thalamo-cortical functional connections during fMRI of human REM sleep, OHBM 2025.
2024.06 Reproducible, data-driven characterization of sleep based on whole-night fMRI. SLEEP Meeting 2024.
2022.08 Effects of sleep duration on neurocognitive development in U.S. early adolescents. Public science talk, University of Maryland, Baltimore.
2022.07 Adolescents' sleep, brain, and behavior. NIAAA, Laboratory of Neuroimaging.
2022.06 Effects of sleep duration on neurocognitive development in U.S. early adolescents. NINDS, Laboratory of Functional and Molecular Imaging.

Professional Affiliations

Organization for Human Brain Mapping, Society for Neuroscience, Sleep Research Society, Sigma Xi

Publications

Manuscripts under review or in preparation (including 3 first-author papers)

1. Yang, F. N., Duyn, J. H., & Xie, W. (2024). Interpreting Health Differences between Self-reported White and Black Children in U.S.: Insights from a Methodological Perspective. *medRxiv*, <https://doi.org/10.1101/2024.10.01.24314712>
2. Yang, F. N.*, Xie, L.*, Galli, O., Detre, J. A., Wolk, D. A., & Rao, H. (2019). Preserved global cerebral blood flow accounts for youthful processing speed in older adults. *bioRxiv*, 665935. <https://www.biorxiv.org/content/10.1101/665935v1.full> (*contributed equally)
3. Picchioni, D., Yang, F. N., de Zwart, J.A., Mandelkow, H., Özbay, P. S., Chen, G., Taylor, P. A., Lam, N., Chappel-Farley, M. G., Chang, C., Liu, J., van Gelderen, P., & Duyn, J. H. (2024). Sleep defined by arousal thresholds reveals decreases in corticocortical functional correlations independently from the conventional sleep stage. *bioRxiv*, <https://doi.org/10.1101/2024.08.09.607376>

4. Zeng, X., Li, Y., **Yang, F. N.**, Del Mauro, G., Yu, J., Lu, R., ... & Wang, Z. (2025). Sleep effects on brain, cognition, and mental health during adolescence are mediated by the glymphatic system. *ArXiv*, arXiv-2512.
5. **Yang, F.N.**, Picchioni, D., & Duyn, J.H. (under review). Deep-Learning Sleep Staging of Whole-Night EEG–fMRI Data Achieves Human-Level Agreement and Aligns With fMRI-Defined Brain States.

Peer-reviewed journal articles (including 14 first-author papers)

1. **Yang, F. N.**, Picchioni, D., de Zwart, J.A., van Gelderen, P., & Duyn, J. H. (2026). What makes our brain unique: insights from a two-night fMRI-EEG sleep study. *Imaging Neuroscience*, 4. <https://doi.org/10.1162/IMAG.a.1333>
2. **Yang, F. N.**, Picchioni, D., de Zwart, J.A., van Gelderen, P., & Duyn, J. H. (2025). Increased intra-thalamic and thalamo-cortical functional connections during fMRI of human REM sleep. *Imaging Neuroscience*, 3. <https://doi.org/10.1162/IMAG.a.1026>
3. **Yang, F. N.**, Picchioni, D., de Zwart, J.A., Wang, Y., van Gelderen, P., & Duyn, J. H. (2024) Reproducible, data-driven characterization of sleep based on brain dynamics and transitions from whole-night fMRI. *eLife*, 13:RP98739. <https://doi.org/10.7554/eLife.98739>
4. **Yang, F.N.**, Picchioni, D., & Duyn, J.H. (2023). Effects of sleep-corrected social jetlag on measures of mental health, cognitive ability, and brain functional connectivity in early adolescence. *Sleep*, zsad259. <https://doi.org/10.1093/sleep/zsad259>
5. **Yang, F.N.**, Liu, T.T., & Wang, Z. Corticostriatal connectivity mediates the reciprocal relationship between sleep and impulsivity in early adolescents. (2023). *Journal of Child Psychology and Psychiatry*, 64(11), 1545–1554. <https://doi.org/10.1111/jcpp.13843>
6. Jiang, X., Dahmani, S., Bronshteyn, M., **Yang, F.N.**, Ryan, J.P., Gallagher Jr, R.C., Damera, S.R., Kumar, P.N., Moore, D.J., Ellis, R.J., (2022). Cingulate transcranial direct current stimulation in adults with HIV. *Plos One*, 17, e0269491.
7. **Yang, F.N. ***, Xie, W. *, & Wang, Z. (2022). Effects of sleep duration on neurocognitive development in U.S. early adolescents: a propensity score matched, longitudinal, observational study. *The Lancet Child & Adolescent Health*, 6(10), 705–712. [https://doi.org/10.1016/S2352-4642\(22\)00188-2](https://doi.org/10.1016/S2352-4642(22)00188-2) (*contributed equally) [*The Cover Article; the Top Publication among all papers used the ABCD data by Altmetric Attention Score = 2511*] [*More than 200 News Mentions and 13,000 Social Media Shares, Likes, & Comments*]
8. **Yang, F.N.**, Liu, T.T., & Wang, Z. (2022). Functional connectome mediates the association between sleep disturbance and mental health in preadolescence. *Human Brain Mapping*, 43(6), 2041–2050. <https://doi.org/10.1002/hbm.25772>
9. **Yang, F.N.**, Hassanzadeh-Behbahani, S.M., Kumar, P., Moore, D.J., Ellis, R.J., Jiang, X. (2022). The impacts of HIV-infection, age, and education on functional brain networks in adults with HIV. *Journal of NeuroVirology*. 28(2), 265–273. <https://doi.org/10.1007/s13365-021-01039-y>
10. Bronshteyn, M., **Yang, F.N.**, Shattuck, K. F., Dawson, M., Kumar, P., Moore, D.J., Ellis, R. J., & Jiang, X. Depression is associated with hippocampal volume loss in adults with HIV. *Human brain mapping*, 42(12), 3750–3759. <https://doi.org/10.1002/hbm.25451>
11. Mao, T., Dinges, D., Deng, Y., Zhao, K., Yang, Z., Lei, H., Fang, Z., **Yang, F.N.**, Galli, O., Goel, N. & Basner, M., (2021). Impaired Vigilant Attention Partly Accounts for Inhibition Control Deficits After Total Sleep Deprivation and Partial Sleep Restriction. *Nature and Science of Sleep*, 13, 1545–1560. <https://doi.org/10.2147/NSS.S314769>
12. **Yang, F.N.**, Hassanzadeh-Behbahani, S., Bronshteyn, M., Dawson, M., Kumar, P., Moore, D. J., Ellis, R. J., & Jiang, X. (2021). Connectome-based prediction of global cognitive performance in people with HIV. *NeuroImage: Clinical*, 30, 102677. <https://doi.org/10.1016/j.nicl.2021.102677>
13. Li, Y., Wang, F., Yan, M., Cantu, E., **Yang, F.N.**, Rao, H., & Feng, R. (2021). Peel Learning for Pathway-Related Outcome Prediction. *Bioinformatics*. 37 (22), 4108–4114. <https://doi.org/10.1093/bioinformatics/btab402>

14. **Yang, F.N.**, Bronshteyn, M., Flowers, S.A., Dawson, M., Diaz, M., Kumar, P., Rebeck, G.W., Turner, R.S., Moore, D.J., Ellis, R.J., & Jiang, X. (2021). Low CD4 nadir exacerbates the impacts of APOE ϵ 4 on memory circuitry in adults with HIV. *AIDS*. 35 (5), 727-736. <https://doi.org/10.1097/QAD.0000000000002840>
15. **Yang, F.N.**, Stanford, M.J., & Jiang, X. (2020). Low cholesterol level linked to reduced semantic fluency performance and reduced gray matter volume in the medial temporal lobe. *Frontiers in Aging Neuroscience*, 12, 57. <https://doi.org/10.3389/fnagi.2020.00057>
16. Chai, Y., Fang, Z., **Yang, F.N.**, Xu, S., Deng, Y., Raine, A., Wang, J., Yu, M., Basner M., Goel, N., Kim, J., Wolk, D.A., Detre, J. A., Dinges, D.F., & Rao, H. (2020). Two nights of recovery sleep restores hippocampal connectivity but not episodic memory after total sleep deprivation. *Scientific Reports*. 10 (1), 1-11. <https://doi.org/10.1038/s41598-020-65086-x>
17. **Yang, F.N.**, Xu, S., Spaeth, A., Galli, O., Zhao, K., Fang, Z., Basner, M., Dinges, D. F., Detre, J.A., & Rao, H. (2019). Test-retest reliability of cerebral blood flow for assessing brain function at rest and during a vigilance task. *NeuroImage*, 193, 157-166. <https://doi.org/10.1016/j.neuroimage.2019.03.016>
18. **Yang, F.N.**, Xu, S., Chai, Y., Basner, M., Dinges, D.F., & Rao, H. (2018). Sleep deprivation enhances inter-stimulus interval effect on vigilant attention performance. *Sleep*, 41(12), zsy189. <https://doi.org/10.1093/sleep/zsy189>
19. Liu, T.T., Nestor, A., Vida, M., Pyles, J., Patterson, C., Yang, Y., **Yang, F.N.**, Freud, E., & Behrmann, M. (2018). Successful reorganization of category-selective visual cortex following occipito-temporal lobectomy in childhood. *Cell Reports*. 24(5), 1113-1122.e6. <https://doi.org/10.1016/j.celrep.2018.06.099>
20. Xu, S., Pan, Y., Qu, Z., Fang, Z., Yang, Z., **Yang, F.N.**, Wang, F., & Rao, H. (2018). Differential effects of real versus hypothetical monetary reward magnitude on risk-taking behavior and brain activity. *Scientific Reports*, 8(1), 3712. <https://doi.org/10.1038/s41598-018-21820-0>
21. **Yang, F.N.**, Pan, S.J., & Li, X. (2016). Beta-adrenoreceptor blockade abolishes atomoxetine-induced Risk Taking, *Physiology & Behavior*, 153, 125-132. <https://doi.org/10.1016/j.physbeh.2015.10.032>
22. Tian, L., Li, X., **Yang, F.**, & Zhao, Y. (2014) The Effect of Novelty Seeking Behavioral Trait on Susceptibility to Drug Addiction and Its Neurobiological Mechanisms, *Advances in Psychological Science*, 22(1), 75-85
23. Zhao, Y., Li, X., Tian, L., & **Yang, F.** (2013). Social Interaction Influence on Adolescent Rodents' Susceptibility to Drug Addiction. *Advances in Psychological Science*, 21(12), 2164–2173